

Qatar-1b

R-2475

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110417 · created 2026-08-23T23:41:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455668.93255 +/- 0.00048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.035
Transit depth (Rp/Rs)^2	3.65 +/- 1.35 [%]
Orbital Inclination (inc)	81.53 +/- 1.75
Airmass coefficient 1 (a1)	0.919 +/- 0.015
Airmass coefficient 2 (a2)	0.056 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.64 %
Best Comparison Star	#3 - [426, 157]
Optimal Method	PSF photometry
Transit Duration (day)	0.05 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.191 ± 0.035 vs 0.1463 (deviation 1.28σ)
Duration fit vs published	0.05 d vs 0.0692 d (deviation 0.96σ)
Mid-time O-C	-1.2 min
Depth SNR	5.5
Residual scatter	1.64 %

Light curve

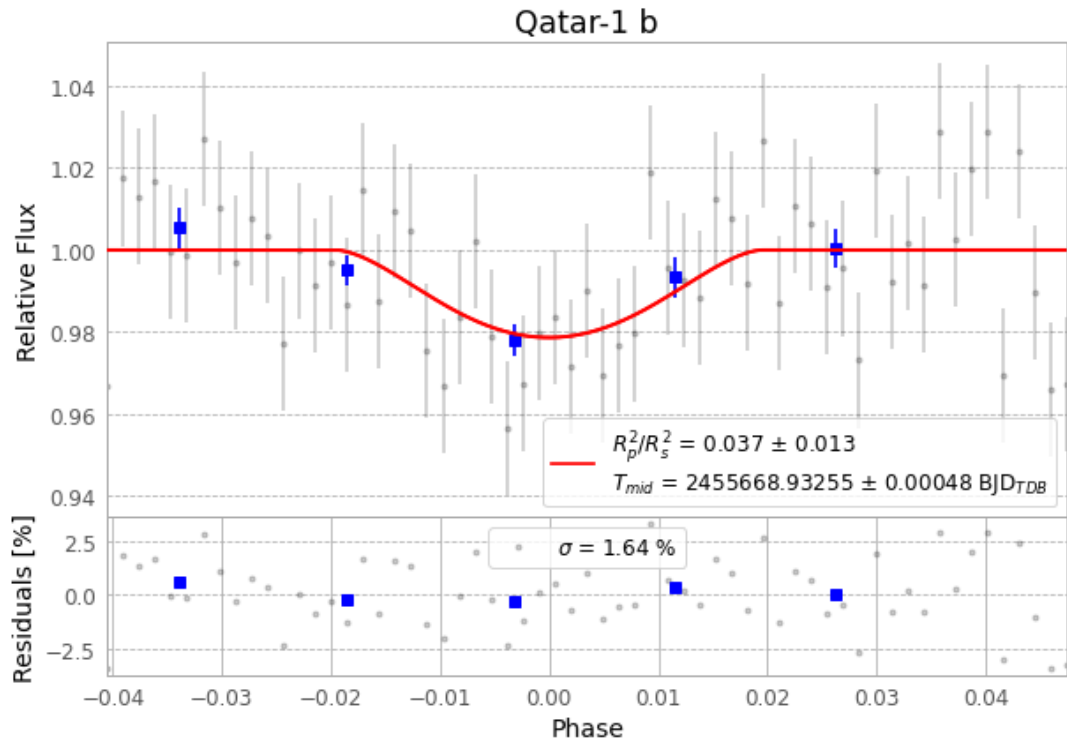


Figure 1 — FinalLightCurve_Qatar-1 b_2011-04-17.png (SHA-256 98ef6d042b932216...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [377, 212], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 acc4ef201f04428f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), Qatar-1110417090013.FITS → Qatar-1110417120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110417.log (e911c00e1534...), FinalLightCurve_Qatar-1 b_2011-04-17.png (98ef6d042b93...), FinalLightCurve_Qatar-1 b_2011-04-17.csv (45963a191d6d...)

Compute resources

Wall time 102.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:41Z.